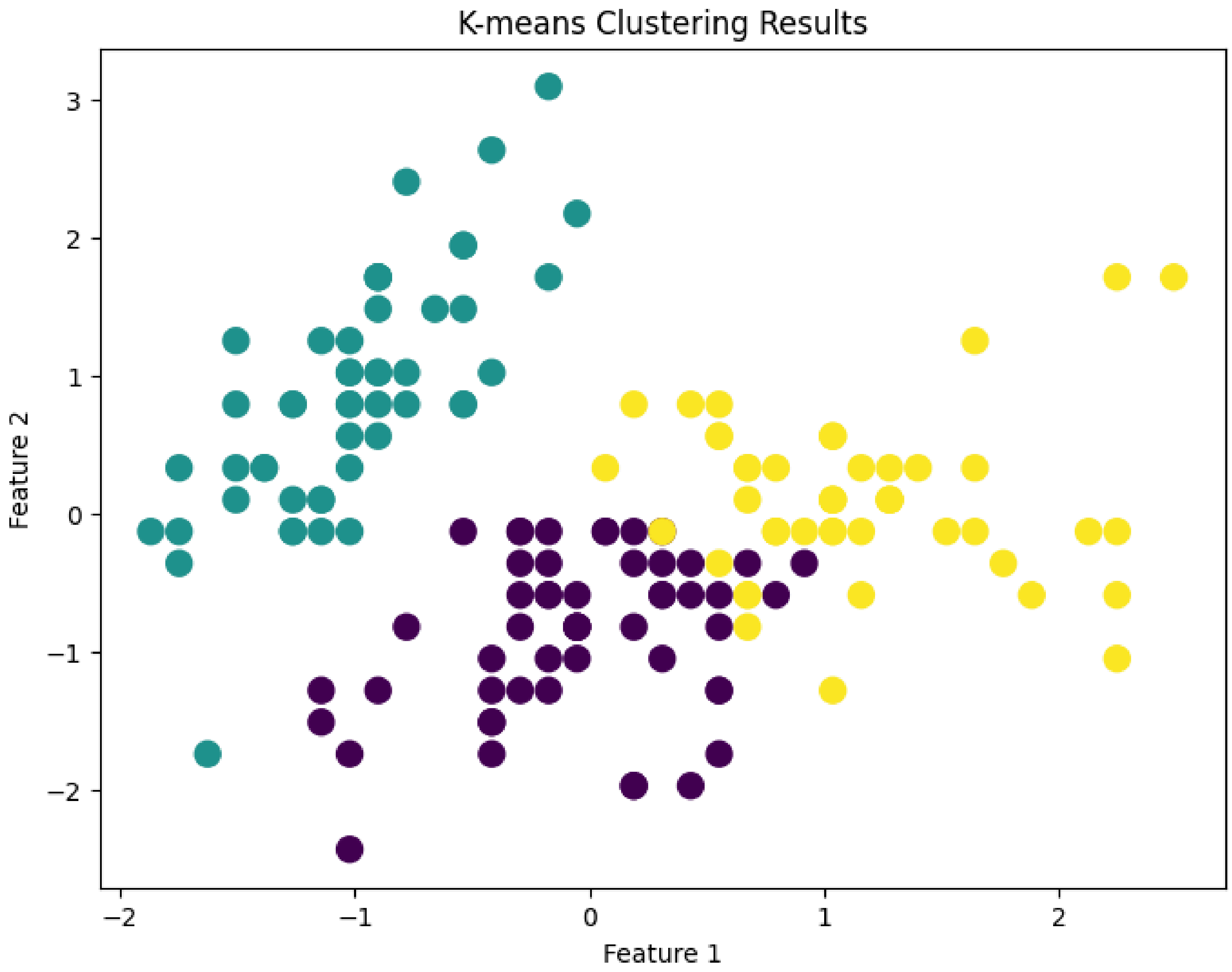
Practical 11

In [3]:
```
import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score, adjusted_rand_score
import matplotlib.pyplot as plt

# 1. Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target

# 2. Preprocess data: Scale the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Apply K-means clustering
kmeans = KMeans(n_clusters=3, random_state=42) # Iris has 3 classes, so we set n_clusters=3
kmeans.fit(X_scaled)

# 4. Get the clustering results
y_pred = kmeans.predict(X_scaled)

# 5. Evaluate the model using Silhouette Score, Inertia, and Adjusted Rand Index
sil_score = silhouette_score(X_scaled, y_pred)
inertia = kmeans.inertia_
ari = adjusted_rand_score(y, y_pred)

# 6. Print the evaluation metrics
print(f"Silhouette Score: {sil_score}")
print(f"Inertia (Sum of squared distances): {inertia}")
```

```python
print(f"Adjusted Rand Index: {ari}")

# 7. Visualize the clustering results (2D projection for simplicity)
plt.figure(figsize=(8, 6))
plt.scatter(X_scaled[:, 0], X_scaled[:, 1], c=y_pred, cmap='viridis', marker='o', s=100)
plt.title('K-means Clustering Results')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.show()

# Optional: Finding the optimal number of clusters (Elbow Method)
inertia_values = []
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    inertia_values.append(kmeans.inertia_)
plt.figure(figsize=(8, 6))
plt.plot(range(1, 11), inertia_values, marker='o')
plt.title('Elbow Method for Optimal K')
plt.xlabel('Number of clusters')
plt.ylabel('Inertia')
plt.show()
```

```
C:\Users\Dell\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\cluster\_kmeans.py:1412: FutureWarnin
g: The default value of `n_init` will change from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppres
s the warning
  super()._check_params_vs_input(X, default_n_init=10)
Silhouette Score: 0.45994823920518635
Inertia (Sum of squared distances): 139.82049635974982
Adjusted Rand Index: 0.6201351808870379
```


K-means Clustering Results
Feature 2
Feature 1
3
2
1
0
−1
−2
−2
−1
0
1
2

```
s the warning
  super()._check_params_vs_input(X, default_n_init=10)
```